

# Information Retrieval

## Relevance Feedback

Real and Assumed

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## Relevance Feedback

Observations:

- **A Query only approximates an information need**
- **Users often start with short queries (poor approximations)**
- **People can improve queries after seeing relevant and non-relevant documents**
  - by adding and removing terms
  - by reweighting terms
  - by adding structure (AND, OR, NOT, PHRASE, etc)

Question: **Can a better query be created automatically by analyzing relevant and non-relevant documents?**

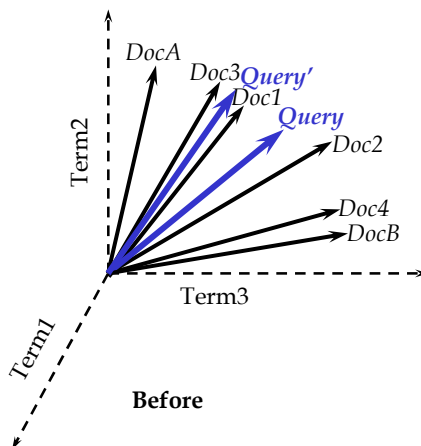
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## Idea appears regularly

- “Real” relevance feedback
  - System returns results
  - User provides some feedback
  - System returns different—better, we hope—results
- “Assumed” relevance feedback
  - System gets results but does not return them
  - Uses returned results to “guess” what was probably meant
  - Modifies query without supervision
  - System returns enhanced—and we hope better—result list
- Occurs in different models
  - Vector space is used most often (we’ll focus on it)
  - Language modeling
    - Excellent success with “assumed” relevance (relevance models)
    - Less obviously good results for “real” feedback

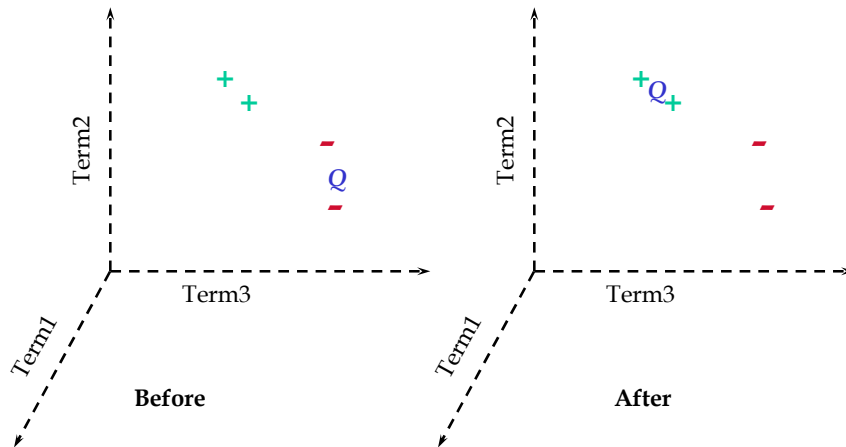
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## Relevance Feedback in the Vector Space



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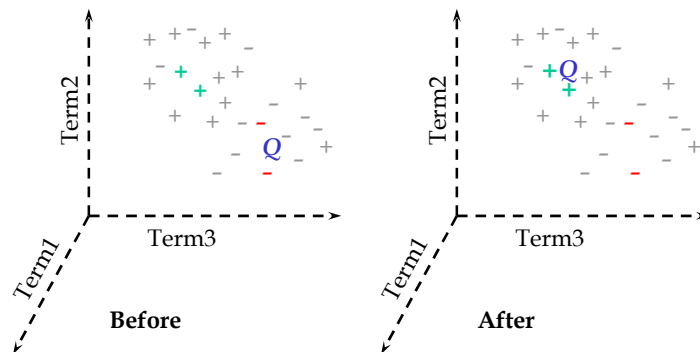
## Same thing as points in space



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## Relevance Feedback in the Vector Space

How can relevance feedback save time if a person has to read documents?



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## Relevance Feedback

Hypothesis: **A better query can be created *automatically* by analyzing relevant and non-relevant documents**

- **Relevant passages and phrases can also be identified, but this is not common**
- **Assumes relevant and non-relevant documents are easy for people to identify**
- **Can be viewed as a form of “query-by-example”**

Common Simplifying Assumptions:

- **Unstructured query (terms and weights, but no operators)**
- **Binary relevance judgements (relevant, not relevant)**

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## Relevance Feedback in the Vector Space

Goal: **Move new query closer to relevant documents**

Approach: **New query is a weighted average of original query, and relevant and non-relevant document vectors**

$$Q' = Q + \alpha \frac{1}{|R|} \sum_{D_j \in R} D_j - \beta \frac{1}{|NR|} \sum_{D_j \in NR} D_j$$

Often written  
 $Q' = \alpha Q + \beta R - \gamma N$

$\alpha$  and  $\beta$  are constants that represent the relative importance of positive and negative feedback

Variations:

- **Different values of  $\alpha$  and  $\beta$**
- **Vector length (number of terms added to the query)**
- **Which documents are used for training**
  - all, best, uncertain, etc

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## Relevance Feedback in the Vector Space:

### Example

Original Query:

**(5, 0, 3, 0, 1)**

Relevance Feedback Formula:

$$Q' = Q + \alpha \frac{1}{|R|} \sum_{D_j \in R} D_j - \beta \frac{1}{|NR|} \sum_{D_j \in NR} D_j$$

Document D1, Relevant:

**(2, 1, 2, 0, 0)**

Document D2, Non-relevant:

**(1, 0, 0, 0, 2)**

$\alpha = 0.50, \beta = 0.25$

**$Q' = Q + 0.5 D1 - 0.25 D2$**

**$= (5, 0, 3, 0, 1) + 0.5 (2, 1, 2, 0, 0) - 0.25 (1, 0, 0, 0, 2)$**

**$= (5.75, 0.50, 4.00, 0.0, 0.5)$**

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## TREC Example: The TREC Topic (An Information Need)

### *Original TREC Topic:*

<num> Number: 106

<dom> Domain: Law and Government

<title> Topic: U.S. Control of Insider Trading

<desc> Description:

Document will report proposed or enacted changes to U.S. laws and regulations designed to prevent insider trading.

<con> Concept(s):

1. insider trading

2. securities law, bill, legislation, regulation, rule

3. Insider Trading Sanctions Act, Insider Trading and Securities Fraud Enforcement Act

4. Securities and Exchange Commission, SEC, Commodity Futures Trading Commission, CFTC, National Association of Securities Dealers, NASD

<fac> Factor(s):

<nat> Nationality: U.S.

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## TREC Example (InQuery): Ad-hoc Query

*Automatically processed query:*

```
#WSUM ( 1.0
!Terms from <title> field:
2.0 #UW50 ( Control of Insider Trading )
2.0 #PHRASE ( #USA Control ) 5.0 #PHRASE ( Insider Trading )
! Terms from <con> field:
2.0 #PHRASE( securities law) 2.0 bill 2.0 legislation 2.0 regulation
2.0 rule 2.0 #3( Insider Trading Sanctions Act)
2.0 #3( Insider Trading and Securities Fraud Enforcement Act )
2.0 #3( Securities and Exchange Commission) 2.0 SEC
2.0 #3(Commodity Futures Trading Commission) 2.0 CFTC
2.0 #3( National Association of Securities Dealers) 2.0 NASD
! Terms from <desc> field:
1.0 proposed 1.0 enacted 1.0 changes 1.0 #PHRASE ( #USA laws )
1.0 regulations 1.0 designed 1.0 prevent
2.0 #NOT(#FOREIGNCOUNTRY) )
```

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## TREC Example: Relevance Feedback Query

*Automatically modified query, top 10 documents judged:*

```
#WSUM (1
3.882349 #UW50( control inside trade) 2.208832 #SUM( #usa control)
145.571381 #SUM( inside trade) 22.084291 #SUM( secure law)
22.693285 bill 20.984898 legislate 10.354733 regulate
6.540223 rule 1.529766 #OD3( inside trade sanction act)
3.290401 #OD4( inside trade secure fraud enforcement act)
4.8404 #OD4( secure exchange commission) 43.578438 sec
0.94752 #OD3( commodity future trade commission) 1.074666 cftc
2.864415 #OD4( national associate secure deal) 21.846081 nasd
0.542252 propose 2.45709 enact 0.988893 change 4.354009 #SUM( #usa
law)
0.799089 design 1.727937 prevent 0.346877 #NOT( #foreigncountry)
4.599784 drexel 2.052418 fine 1.845434 subcommittee
1.69074 surveillance 1.597542 markey 1.528179 senate
1.186563 manipulate 1.101982 pass 1.060453 scandal
0.921561 edward )
```

## Relevance Feedback in the Vector Space: Variations

Term Selection:

- None (original query terms, only)
- All terms
- Most common terms
- Most highly weighted terms

Weighting:

- **Ide** :  $\alpha=1$ ,  $\beta=1$ , don't normalize by number of judged documents
- **Ide Dec Hi**:  $\alpha=1$ ,  $\beta=1$ , use only the highest ranked non-relevant document(s), don't normalize by number of judged documents
- **Rocchio**: Choose  $\alpha$  and  $\beta$  such that  $\alpha > \beta$  and  $\alpha + \beta = 1$

$$Q' = Q + \alpha \frac{1}{|R|} \sum_{D_j \in R} D_j - \beta \frac{1}{|NR|} \sum_{D_j \in NR} D_j$$

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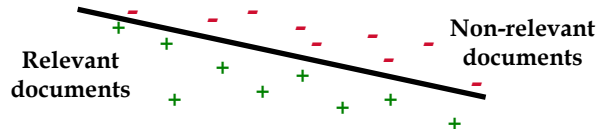
## Relevance Feedback in the Vector Space: Common Choices

- Ide Dec Hi is effective when there are *a few* judged documents
- Rocchio ( $\alpha=0.75$ ,  $\beta=0.25$ ) is effective when there are *many* judged documents
- Expanding by *all* terms is best, but selecting *most common* terms also works well
  - Depends somewhat on the retrieval model
- Coping with negatively weighted terms
  - Vector space does not allow negative weights for cosine similarity
  - Usually drop terms that end up negatively weighted
  - Can create a “not like this” vector consisting of negative terms
    - Difficult to balance issues correctly

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## Relevance Feedback: Machine Learning Approaches

- An unstructured vector query is a linear discriminator
  - e.g.,  $w_1 * t_1 + w_2 * t_2 + \dots + w_n * t_n$
- The goal is to learn weights that separate the relevant documents from the non-relevant documents



- If the documents are *linearly separable*, a learning algorithm can be chosen that is guaranteed to converge to an optimal query
- If the documents are not linearly separable, a learning algorithm can be chosen that minimizes the total amount of error

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## Relevance Feedback: Machine Learning Approaches

- Unstructured queries:
  - Perceptron algorithm (Rocchio)
  - EG (a form of Perceptron algorithm)
  - Regression
  - Neural network algorithms
  - SVM
  - : : :
- Structured queries
  - Decision trees
  - Neural network algorithms
  - Sleeping Experts
  - Ripper
  - : : :

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## Rocchio and the Perceptron

- The Rocchio relevance feedback algorithm is similar to the fixed increment version of the Perceptron rule:

$$\bar{Q}' = \bar{Q} \pm c\bar{D}_i \quad \begin{cases} + & \text{if } D_i \in R \\ - & \text{if } D_i \in \bar{R} \end{cases}$$

- The Perceptron:
  - requires *repeated* exposure to training data,
  - requires random sampling,
  - works best if R and NR are of similar size, and
  - is optimal if R and NR can be separated by a hyperplane (otherwise it oscillates).

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## Relevance Feedback: Simple Methods for Adding Structure

Basic Process:

- Generate candidate operators (Boolean, Phrase, proximity, etc)
  - algorithms: exhaustive, greedy/selective
- Add some or all candidates to document representations
- Weight like other terms

Effectiveness:

- Extremely effective for proximity operators
- Boolean?

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## Relevance Feedback

- Relevance Feedback could also modify *document* representation
  - document space modification
  - connectionist learning (changing weights in network)
- Assumptions:
  - a person's relevance judgements are consistent
  - modifications for one person are meaningful for another
- Never shown to be effective *consistently*
- An old idea, periodically resurfaces
  - recommender systems
- Difficult to figure out how searchers should use it

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## Relevance Feedback: Summary

- Relevance feedback can be very effective
- Effectiveness depends on number of judged documents
- Significantly outperforms best human queries, given enough judged documents
- Results can be unpredictable with less than five judged documents
- Not used often in production systems, e.g., Web
  - consistent mediocre performance preferred to inconsistently good/great results
  - Stick with “documents like this one” variant
- An area of very active research (many open questions)

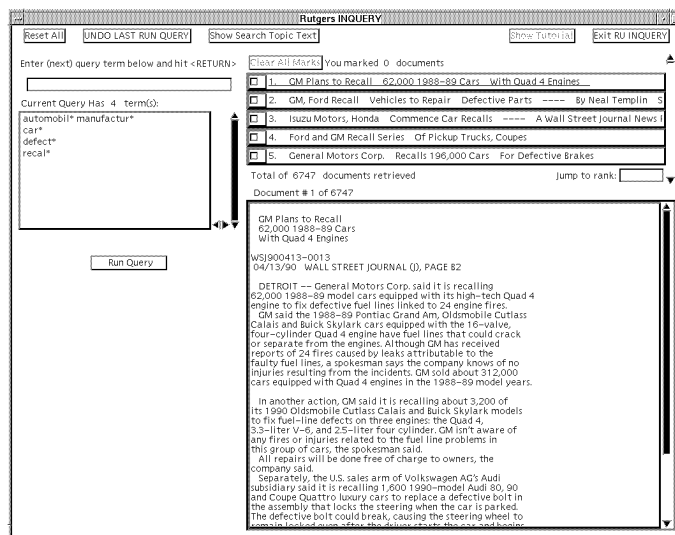
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## Relevance Feedback: Using it

- Relevance feedback is not widely used
- Few studies explore the user side of feedback
  - Don't necessarily answer that question, but are still interesting
- Jürgen Koenemann and Nick Belkin looked at this
  - “A case for interaction: A Study of Interactive Information Retrieval Behavior and Effectiveness”, CHI 1996
- User study of 64 users
- Presented with three styles of relevance feedback
  - Opaque, relevance feedback is “magic” behind the scenes
  - Transparent, same as *opaque* but users shown expansion terms
  - Penetrable, user given chance to edit list of terms before re-run

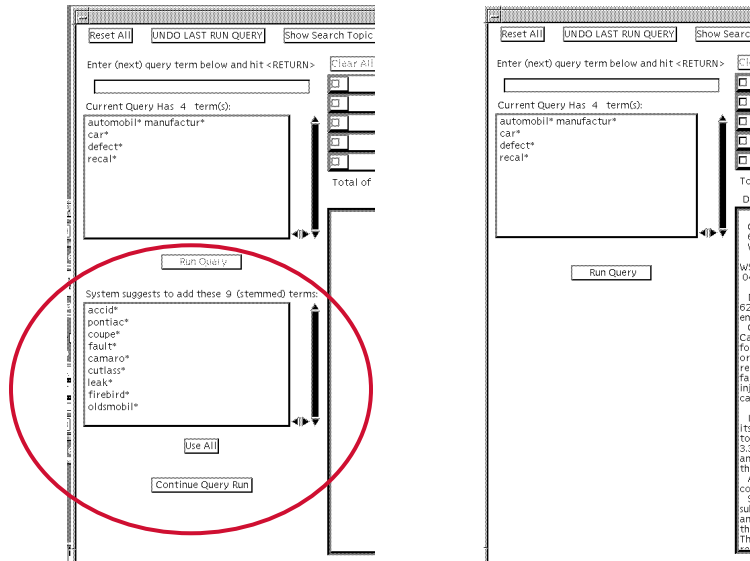
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## Base system used (no feedback)



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## Allowing user access (penetrable)



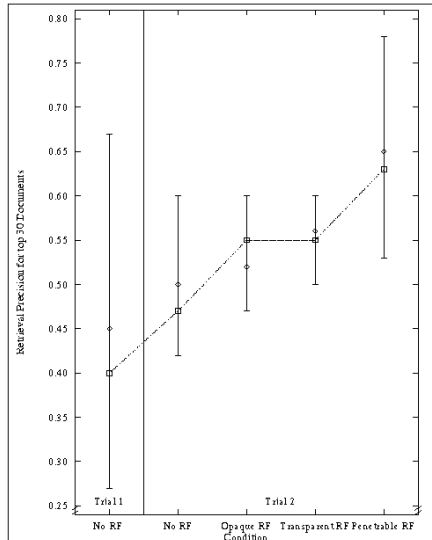
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## Experimental procedure

- Two query construction approaches
  - First without relevance feedback
  - Second with one of three RF approaches (randomly assigned)
- Task is to construct a good long-term query
- Evaluation is based on effectiveness of final query
- No difference between users on first task

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## Effectiveness with Feedback

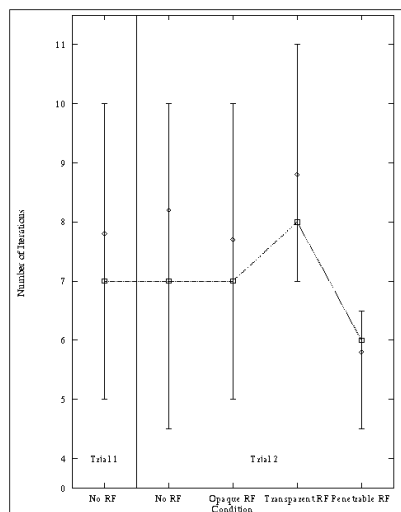


- Precision at 30 documents
- Clear improvements from use of RF
- Opaque and transparent the same (by design)
- Penetrable best
- Only statistically significant difference is between penetrable and base
- Results comparable for precision at 100 documents

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## Behavior with Feedback

- Task was to build a good query
- How many attempts do people make?
- For some reason, transparent interface encouraged an extra iteration
- Penetrable interface took one less than "normal"
- Not clear what this means



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## Was Feedback Used by Searcher?

- Where did words they chose come from?

- Copied from lists provided by feedback
- Added automatically by system

- Users typed short queries
- Feedback added many terms
- Penetrable system encouraged fewer terms
  - But resulted in more effective queries (faster)

| Mean Number & Sources of Query Terms |                 |                         |      |                 |      |
|--------------------------------------|-----------------|-------------------------|------|-----------------|------|
| Relevance Feedback Condition         | User User Typed | Controlled Copy from RF | Σ    | Added by RF SYS | Σ    |
| <b>Topic 162:</b>                    |                 |                         |      |                 |      |
| None                                 | 6.9             | n/a                     | 6.9  | n/a             | 6.9  |
| Opaque                               | 10.9            | n/a                     | 10.9 | 35.9            | 46.8 |
| Transparent                          | 3.3             | 9.1                     | 12.4 | 42.8            | 55.1 |
| Penetrable                           | 6.3             | 24.4                    | 30.6 | n/a             | 30.6 |
| <b>Topic 165:</b>                    |                 |                         |      |                 |      |
| None                                 | 6.0             | n/a                     | 6.0  | n/a             | 6.0  |
| Opaque                               | 3.8             | n/a                     | 3.8  | 20.5            | 24.3 |
| Transparent                          | 4.3             | 5.3                     | 9.5  | 17.8            | 27.3 |
| Penetrable                           | 3.3             | 9.5                     | 12.8 | n/a             | 12.8 |
| <b>162&amp;165:</b>                  |                 |                         |      |                 |      |
| None                                 | 6.4             | n/a                     | 6.4  | n/a             | 6.4  |
| Opaque                               | 7.3             | n/a                     | 7.3  | 28.2            | 35.5 |
| Transparent                          | 3.8             | 7.2                     | 10.9 | 30.3            | 41.2 |
| Penetrable                           | 4.8             | 16.9                    | 21.7 | n/a             | 21.7 |

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## Subjective reactions

- Subjects “liked” the penetrable version
- Subjects in opaque condition expressed desire to “see and control” what happened
- Subjects comments that feedback made them “lazy”
  - Task of generating terms changed to task of selecting terms

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## Relevance Feedback: Assumed

- True relevance feedback is supervised
  - Feedback is done based on *genuine* user annotations
- What happens if we try to guess what is relevant?
- Assume many top ranked documents are relevant
  - Optionally find a collection of probably non-relevant documents
- Modify query on that assumption
- Re-run that new query and show results to user
- What happens?
- Pseudo-relevance feedback
  - Blind relevance feedback
  - Local feedback
  - ...

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## History

- Attar and Fraenkel, JACM 1977
  - “Local Feedback in Full-Text Retrieval Systems”
  - Small databases
    - 75 US patents in electronics, 7 queries
    - Some work with 2500 documents in Hebrew
  - Showed clear gain from “local dynamic feedback”
  - Helped to weight candidate terms by proximity
- Idea seems to have been largely ignored for 16 years
  - In published results, at least...

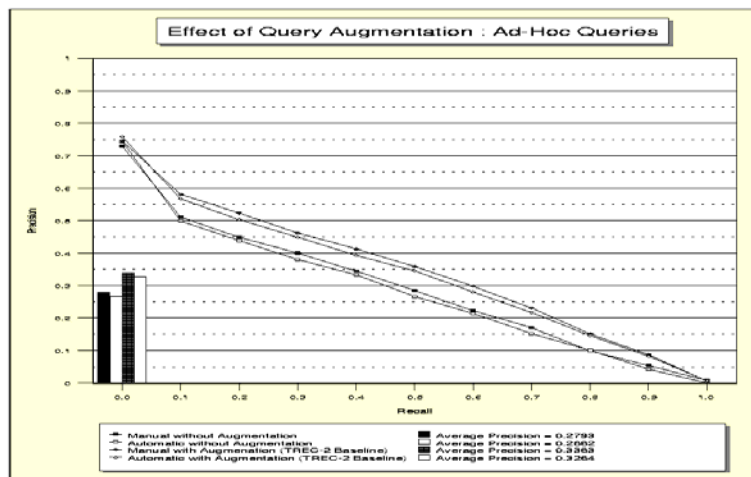
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## History, modern

- Rediscovered in TREC-2 (1993)
- Robertson, Walker, et al, City University of London
  - “Okapi at TREC-2”
- (Also done by UCLA and CMU/Claritech)
- Procedure
  - Used top 30 retrieved documents
  - Added as many terms as half the length of the query
    - TREC-2 had very long queries (30-50 words)
- Example, topic 112 “funding biotechnology”
  - Added *pharmaceut, financ, partner, drug, investor, technologi, company, industri, and develop.*
- Results
  - Average precision went from 37.3% to 40.7%
  - Not huge, but noticeable in an environment like TREC
  - NOTE: 25 queries improved, 18 got worse, and 7 didn't change

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## Example of impact



Evans and Lefferts, “Design and Evaluation of the CLARIT TREC-2 System”

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## History, continued

- Widely adopted in TREC-3
- “Okapi at TREC-3” (Robertson et al)
  - Added up to 40 terms (averaging around 20 additions)
- “Automatic query expansion using Smart” (Buckley et al)
  - Massive expansion: 500 terms and 10 phrases!
- “TREC-2 .. Using PIRCS” (Kwok et al)
  - Spreading activation triggered by six top-ranked documents
  - On average added 11 new terms
- UMass Amherst CIIR’s run still a hold-out
  - Used a global type of query expansion (InFinder)
    - Attar and Fraenkel compared their local feedback technique to such an approach
  - By TREC-5 had “local context analysis” as a passage-focused variation on expansion
- Continues in TREC-4, but new ranking function dominates
  - (The “Okapi tf” function appeared in TREC-3.)

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## Why in 1993, but not 1977?

- Attar and Fraenkel showed suggestive results
- But techniques generally went awry
  - Still did in 1993, but helped more than it hurt
  - Standard class project in some IR classes in 1980s: didn’t work well
- Nature of TREC collections different
  - More relevant documents per query, increases chance of hit
  - Better retrieval algorithms, also increases change of hit
  - Longer documents, so more term usage to help find good ones
- Still widely used
  - Local context analysis (LCA) format used particularly at CIIR
  - Language modeling approaches incorporate similar ideas as part of smoothing technique
  - Also explicitly considered as part of relevance models

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## Local Context Analysis (LCA)

[Xu and Croft, 1996]

- Assumed relevance feedback
- Observations
  - Existing techniques improved queries on average
  - But some queries had serious drop in effectiveness
  - Top ranked documents were not always right
  - Often caused by match of a single query word
  - Not every word is useful to add to queries
- Inspired creation of LCA
- Major focus is on getting better terms for expansion
  - Finding terms to consider
  - Selection of terms
  - Weighting of selected terms

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## Finding candidate terms

- Run query to retrieve passages
  - Similar to most “assumed” relevance work
  - Passage-retrieval unique to LCA (at the time)
  - Uses 300-word passages
- Select expansion concepts from retrieved set
- Why passages?
  - Minimizes spurious concepts that occur in lengthy documents

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## Selecting candidate terms

- Parse document collection
- Generate part of speech tagging
  - The/AT **bill/NN** has/HVZ been/BEN reworked/VBN since/CS it/PPS was/BEDZ introduced/VBN ./, in/IN **order/NN** to/TO meet/VB some/DTI **employer/NN objections/NNS** ./ But/CC the/AT **measure/NN** still/RB is/BEZ opposed/VBN by/IN the/AT **construction/NN industry/NN** ./, which/WDT argues/VBZ that/CS it/PPS would/MD impose/VB **unionism/NN** and/CC higher/JJ **costs/NNS** on/IN much/AP of/IN the/AT **industry/NN** 's/\$ **work/NN** ./
- Select only noun phrases
  - Shown to be critical in most retrieval systems
  - Generally particularly useful for expansion
  - Could easily be extended if useful
    - Adjective-noun phrases, verbs, ...
  - Note that tagging is automated, so makes mistakes!

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## Weighting terms

- Want “concepts” that occur near query words
  - The more query words they occur near, the better
  - Count co-occurrences in 300-word windows of text (passages)
    - To avoid coincidental co-occurrence in a large document
- Uses the following ad-hoc function to weight concepts

$$f(c, Q) = \prod_{w_i \in Q} (0.01 + \text{co\_degree}(c, w_i))^{\text{idf}(w_i)}$$

$$\text{co\_degree}(c, w) = \max\left(\frac{n_{cw} - \text{En}(c, w) - 1}{n_c}, 0\right)$$

$$\text{En}(c, w) = \frac{n_w n_c}{N}$$

$$\text{idf}(w) = \min(1.0, \log_{10}(N/n_w)/5)$$

Importance of word

Measure co-occurrence

Floor the IDF component

Slow its growth

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## Using expanded query

- Developed using Inquiry
- Incorporate using weighted sum
  - Weight original query and expansion query equally

$$Q_{\text{new}} = \#wsum( 1.0 \ 1.0 \ Q_{\text{original}} \\ 1.0 \ Q_{\text{Ica}} )$$

$$Q_{\text{Ica}} = \#wsum( 1.0 \ 1.0 \ c_1 \ 1.0 \ c_2 \ \dots \ 1.0 \ c_{30} )$$

- Variations
  - Lower weight on each subsequent term
    - More important the more terms that are added
  - Weight original query equally with a single expansion concept
    - Only works when query is not very reliable

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## Example of expansion concepts

- TREC query 213
  - As a result of DNA testing, are more defendants being absolved or convicted of crimes?
- Expansion concepts
 

|                                           |                         |                           |                     |
|-------------------------------------------|-------------------------|---------------------------|---------------------|
| dna-pattern                               | dna-testing             | lifecodes                 | dna-test-results    |
| dna-lab                                   | dna-evidence            | dna-test                  | dna-profile         |
| defense-attorneys-challenging-reliability |                         |                           | bureau-expert       |
| lawyer-peter-neufeld                      |                         | new-york-city-murder-case |                     |
| michael-baird                             | procedures-track-record | dna-laboratory            |                     |
| oregon-rape-case                          | mark-storolow           | laboratory-geletin        |                     |
| supermarket-merchandise                   |                         | thomas-caskey             | procedures-lifecode |
| lifecodes-corp                            | tests-reliability       | maine-case                | rape-conviction     |
| dna-strand                                |                         |                           |                     |

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## Does it work?

- TREC-3 and TREC-4 ad-hoc queries
- With and without LCA expansion

Precision (% change) – 50 queries

| Recall  | baseline | corpus-query |
|---------|----------|--------------|
| 0       | 82.2     | 85.3 (+3.8)  |
| 10      | 57.3     | 65.1 (+13.5) |
| 20      | 46.2     | 54.7 (+18.5) |
| 30      | 39.1     | 46.8 (+19.9) |
| 40      | 32.7     | 40.0 (+22.1) |
| 50      | 27.5     | 34.6 (+25.9) |
| 60      | 22.6     | 28.4 (+25.2) |
| 70      | 18.0     | 23.0 (+27.3) |
| 80      | 13.3     | 17.4 (+30.7) |
| 90      | 7.9      | 10.7 (+34.4) |
| 100     | 0.5      | 0.7 (+36.9)  |
| average | 31.6     | 37.0 (+17.0) |

TREC-3

Precision (% change) – 49 queries

| Recall  | baseline | corpus-query |
|---------|----------|--------------|
| 0       | 71.0     | 70.4 (−0.8)  |
| 10      | 49.3     | 54.3 (+10.0) |
| 20      | 40.4     | 45.0 (+11.6) |
| 30      | 33.3     | 37.7 (+13.4) |
| 40      | 27.3     | 32.6 (+19.4) |
| 50      | 21.6     | 27.4 (+26.7) |
| 60      | 14.8     | 20.8 (+40.7) |
| 70      | 9.5      | 13.6 (+43.3) |
| 80      | 6.2      | 8.2 (+33.8)  |
| 90      | 3.1      | 4.2 (+34.2)  |
| 100     | 0.4      | 0.6 (+36.7)  |
| average | 25.2     | 28.6 (+13.7) |

TREC-4

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## Summary

- Relevance feedback
  - Real or assumed
- Real relevance feedback
  - Usually improves effectiveness significantly
  - Not always stable with very few documents judged
  - Difficult to incorporate into a usable system
  - “Documents like this one” is a simple instance
- Assumed relevance feedback
  - Also called “pseudo relevance feedback” or “local feedback”
    - Or “quasi-relevance feedback” or ...
  - Rocchio-based approaches effective but unstable
  - LCA comparably effective (maybe better) but more stable
  - Relevance models provide formal framework

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